Marion L'HERMITE

Biomedical Engineer

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Education

2025-2026 **National Technical University of Athens**

- Research Methodology
- Life Sciences for Engineering
- Biostatistics and Machine Learning
- Biosignal Acquisition and Processing
- Biomechanics

2023-2025 **Institut Polytechnique de Paris - Télécom SudParis**, Generalist Engineering Program

- Mathematics, Statistics, Optimization and Data Analysis
- Multimedia acquisition, coding and compression, Visual Content Analysis reality
- Computer Science and Information Systems
- Microwave and Electronics
- Signal and communication theory

2021-2023 **Preparatory Classes MPSI then MP**, with a specialization in Industrial Sciences

Intensive training in mathematics, physics, and engineering sciences.

2020 **Scientific Baccalaureate**, with Highest Honors, with a specialization in Mathematics

Professional Experience

Apr - Oct 2026 **Flomics Biotech — Internship in Bioinformatics**

- Developed bioinformatics and machine learning pipelines for **cfRNA-seq analysis** and the prediction of biological age from high-dimensional transcriptomic data.
- Processed and analyzed transcriptomic datasets using **Python and R**, including data preprocessing, quality control, normalization, statistical analysis, and feature selection.
- Evaluated **regression and tree-based machine learning models**.
- Developed **biology-driven feature engineering** strategies based on gene co-expression and biological pathways to reduce dimensionality and improve model robustness and interpretability.

Projects

Jan - May 2025 **Project in collaboration with the Genoscope (French National Sequencing Center)**

Developed an automated workflow under the supervision of the Genoscope research team to support the LAGE project, which investigates global marine eukaryotic diversity. The workflow grouped contigs into MAGs, using Answer Set Programming-based rules and optimization criteria.

Extracurricular Activities

2024-2025 **Annual Conference – Télécom SudParis**

- Managed and hosted a conference, aimed at introducing students to IoT concepts through real-world applications.
- Recruited and engaged expert speakers from both academia and industry, shaping their contributions to meet educational goals.

2024-2025 **Vice-President – Arts Office of Télécom SudParis**

Oversaw a 40-member team in organizing cultural events, managing logistical, administrative, and budgetary challenges to ensure smooth execution.

Skills

Programming Languages: Python, C, Java, JavaScript, HTML/CSS, PHP, R, Bash, SQL
Frameworks & Libraries: OpenCV, TensorFlow, PyTorch, Pandas
Software and Tools: Git, Autodesk Fusion 360, Microsoft Excel
Languages: French (Native), English (C1), Spanish (B1)